Trust-Aware Multi-Agent Exploration

Problem

Multi-robot exploration of a hidden reward map where robots collect and communicate observations with a central planner. A subset of the robots may be adversarial and report incorrect reward values.

The planner must efficiently explore the environment and reconstruct the reward map from the received reports, while identifying and reducing the influence of adversarial information through trust learned from accumulated cross-check evidence.

Methodology

Adversarial Behaviour

Adversarial robots generate reports with a spatially coordinated but intermittent bias. For a reported cell s, the direction of the bias is determined by the cell location:

$$\text{sign}(s) = \begin{cases} -1, & \text{if hash}(s) \bmod 2 = 0, \\ +1, & \text{otherwise.} \end{cases}$$

Thus, each cell has a fixed bias direction, causing adversarial robots to exhibit a consistent spatial pattern rather than independently choosing the direction of the lie at each observation.

The magnitude of the adversarial bias is sampled as

$$b \sim \mathcal{U}(0.15, 0.30).$$

Since the contradiction threshold is 0.2, biases below this threshold may not produce an observable contradiction, while larger biases can generate detectable disagreements. Consequently, adversarial robots do not produce continuously obvious lies; their reports can appear consistent with honest observations at some cells while becoming contradictory at others.

The reported value by an adversarial robot is given by

$$G'(s) = G(s) + \text{sign}(s)\, b + \epsilon, \qquad b \sim \mathcal{U}(0.15, 0.30), \quad \epsilon \sim \mathcal{N}(0, 0.05).$$

Here, $G(s)$ denotes the true value of cell s, $G'(s)$ denotes the value reported by the adversarial robot, $\text{sign}(s)$ determines the spatially consistent direction of the bias, b is the bias magnitude, and ϵ is the observation noise.

Trust Estimation

Trust weights α_i are estimated from the cross-check history of each agent:

$$\alpha_i = f_\theta(c_i, z_i, D_i),$$

where c_i is the number of contradictions observed for agent i and z_i is the standardized contradiction count,

$$z_i = \frac{c_i - \mu_c}{\sigma_c + \epsilon}$$

For each cell s reported by agent i, the residual disagreement is computed against all other agents j that also reported an observation for that cell:

$$D_i = \frac{1}{|\mathcal{S}_i|} \sum_{s \in \mathcal{S}_i} \frac{1}{|\mathcal{J}_i(s)|} \sum_{j \in \mathcal{J}_i(s)} \left| r_i(s) - r_j(s) \right|,$$

where $\mathcal{S}_i$ is the set of cells reported by agent i, and $\mathcal{J}_i(s)$ is the set of other agents that reported an observation for cell s.

Map Reconstruction & Objective

The estimated map $M(s)$ at state s is reconstructed as the trust-weighted mean:

$$M(s) = \frac{\sum_{i \in R_s} \alpha_i \, r_i(s)}{\sum_{i \in R_s} \alpha_i}$$

The network parameters θ_f are optimised directly through reconstruction loss:

$$\min_{\theta_f} \mathcal{L} = (G(s) - M(s))^2$$

Since reported values $r_i(s)$ are fixed, gradients propagate solely through the weighting factors α_i, driving separation between honest and adversarial agents.

Current Understanding

- As multiple robots visit overlapping cells, evidence accumulates. Cross-check counts increase:

  $$c_{\text{liar}} \uparrow \quad \Longrightarrow \quad \alpha_{\text{honest}} \gg \alpha_{\text{liar}}$$

- As the trust model converges, the relative ratio between trust values becomes stable. Therefore, we should not expect $\alpha_{\text{honest}} \to 1$ and $\alpha_{\text{liar}} \to 0$.
- If a cell has only a single observation and that observation comes from an adversarial robot, its information becomes unreliable as

  $$\alpha_{\text{liar}} \to 0.$$

  Such cells should become candidates for revisiting and be prioritised by the planner.

Thoughts - Planner

Planner Value Map

Let $M_t(s)$ be the trust-weighted reconstruction at cell s at time t (as defined above), $U_t(s) = \sum_{i \in R_s} \alpha_i$ the trusted support, and $N_t(s)$ the visit count. Define the state maps

$$X_t = \big(U_t,\, M_t,\, N_t\big).$$

A CNN g_ϕ maps the full grid state to a raw value map:

$$R_t = g_\phi(X_t) \in \mathbb{R}^{grid_size \times grid_size}.$$

$R_t(s)$ is the predicted value of visiting cell s next.

Target: Realised Information Gain (Local)

For any cell s visited at step t, define the realized error drop

$$y_t(s) = \big(G(s) - M_t(s)\big)^2 - \big(G(s) - M_{t+1}(s)\big)^2.$$

$y_t(s)$ is only defined for visited cells; unvisited cells are excluded from training.

Training g_ϕ

$$\min_\phi \; \mathcal{L}_t = \frac{1}{|\mathcal{V}_t|} \sum_{s \in \mathcal{V}_t} \big(R_t(s) - y_t(s)\big)^2,$$

where $\mathcal{V}_t$ is the set of cells visited at step t.

The raw value map R_t can be used directly as input to a downstream planner. It estimates how much the global reconstruction error is expected to decrease if a reliable observation is obtained at each cell s.

For validation, the predicted value map R_t can be compared with the oracle value map Y_t by converting both maps into probability distributions. The oracle probability distribution is defined as

$$P_t^*(s) = \frac{Y_t(s)}{\sum_{u \in \mathcal{G}} Y_t(u)},$$

while the predicted probability distribution is

$$P_t(s) = \frac{R_t(s)}{\sum_{u \in \mathcal{G}} R_t(u)}$$

The similarity between the two distributions can then be measured using the KL divergence:

$$D_{\mathrm{KL}}(P_t^* \| P_t) = \sum_{s \in \mathcal{G}} P_t^*(s) \log\left(\frac{P_t^*(s)}{P_t(s)}\right).$$

A lower KL divergence indicates that the predicted value distribution is closer to the oracle distribution, with $D_{\mathrm{KL}} = 0$ indicating identical distributions.

Analytic Derivation

Let $M_t(s)$ denote the current estimate of cell s at time t. For all agents that have reported an observation for cell s, define the trust-weighted observation sum as

$$W_t(s) = \sum_i \alpha_i r_i(s),$$

and the total trust value of all agents that reported cell s as

$$U_t(s) = \sum_i \alpha_i.$$

Therefore, the current trust-weighted estimate of the cell is

$$M_t(s) = \frac{W_t(s)}{U_t(s)}.$$

Let $T(s)$ denote the ground-truth reward of cell s, which is unknown to the central planner. The current squared estimation error is therefore

$$e_t^2(s) = (T(s) - M_t(s))^2 .$$

Now suppose an honest robot is sent to cell s at time $t + 1$. Since the robot is assumed to be honest, its observation is

$$r_{\mathrm{new}}(s) = T(s), \qquad \alpha_{\mathrm{new}} = 1.$$

The updated trust-weighted observation sum and total trust become

$$W_{t+1}(s) = W_t(s) + T(s), \qquad U_{t+1}(s) = U_t(s) + 1.$$

Hence, the expected updated estimate after obtaining this honest observation is

$$M_{t+1}(s) = \frac{W_t(s) + T(s)}{U_t(s) + 1}.$$

Using

$$W_t(s) = U_t(s)M_t(s),$$

this can equivalently be written as

$$M_{t+1}(s) = \frac{U_t(s)M_t(s) + T(s)}{U_t(s) + 1}.$$

Therefore, the new estimation error is

$$\begin{aligned} e_{t+1}(s) &= T(s) - M_{t+1}(s) \\ &= T(s) - \frac{U_t(s)M_t(s) + T(s)}{U_t(s) + 1} \\ &= \frac{T(s)U_t(s) + T(s) - U_t(s)M_t(s) - T(s)}{U_t(s) + 1} \\ &= \frac{U_t(s)}{U_t(s) + 1}\left(T(s) - M_t(s)\right) \\ &= \frac{U_t(s)}{U_t(s) + 1} e_t(s). \end{aligned}$$

Hence, the new squared error is

$$e_{t+1}^2(s) = e_t^2(s)\frac{U_t(s)^2}{(U_t(s) + 1)^2}.$$

The reduction in squared error, which corresponds to the information-gain target $y_t(s)$, is therefore

$$\begin{aligned} y_t(s) &= e_t^2(s) - e_{t+1}^2(s) \\ &= e_t^2(s)\left(1 - \frac{U_t(s)^2}{(U_t(s) + 1)^2}\right) \\ &= e_t^2(s)\left(\frac{U_t(s)^2 + 2U_t(s) + 1 - U_t(s)^2}{(U_t(s) + 1)^2}\right) \\ &= \boxed{e_t^2(s)\frac{2U_t(s) + 1}{(U_t(s) + 1)^2}} \end{aligned}$$

Thus, before deciding which cell to visit, the effect of obtaining one additional honest observation can be evaluated analytically from the current squared error and the existing trust mass $U_t(s)$.

Why Analytic Derivation?

The information gain from an additional honest observation at cell s factors as

$$y_t(s) = e_t^2(s) \underbrace{\frac{2U_t(s) + 1}{(U_t(s) + 1)^2}}_{\text{known analytically}}.$$

The factor $e_t^2(s) = (T(s) - M_t(s))^2$ is unknown, since $T(s)$ is not available to the planner. The factor $\frac{2U_t(s)+1}{(U_t(s)+1)^2}$ depends only on the trust mass $U_t(s)$, which the planner already knows, so it need not be learned.

Rather than training g_ϕ to approximate the full product $y_t(s)$ – forcing it to implicitly rediscover this known closed-form relationship – we let it learn only the unknown error term:

$$\hat{e}_t^2(s) = g_\phi(X_t), \qquad \boxed{\widehat{y}_t(s) = \hat{e}_t^2(s)\,\frac{2U_t(s) + 1}{(U_t(s) + 1)^2}}$$

This narrows the learning target to genuinely unknown information, while the known analytic effect of trust mass is applied directly.

Decentralised System

The next upgrade to this system is to remove the central planner entirely. Currently, one central node sees everything, all reports from all agents and uses this to compute trust α_i, reconstruct $M_t(s)$, and predict R_t. To decentralise this, every agent must be able to do the same computation locally, which means every agent needs (almost) all the information.

The difficulty is communication. If agents i, j, k meet pairwise, then after i and j communicate, i knows everything j knows (including what j heard from k). Later, when i and k communicate, k ends up with most of what j had too. So information mixes quickly, but naively exchanging *everything* at every meeting is wasteful: most of it is usually already shared, and only a small part is new. This is a synchronisation problem; efficiently finding and sending only the difference between two large, mostly-overlapping datasets. We **can** solve it using a **Merkle tree**, a structure widely used in cryptography for exactly this purpose.

Setup. Each agent i maintains an observation set

$$O_i = \{ o_1, \ldots, o_n \}, \qquad o_k = (s_k,\, r_i(s_k),\, \tau_k),$$

and builds a binary tree of hashes over it:

$$h(\text{leaf}_k) = \text{hash}(o_k), \qquad h(\text{node}) = \text{hash}\big(h(\text{left}) \,\|\, h(\text{right})\big),$$

up to a single root hash H_i that summarises all of O_i.

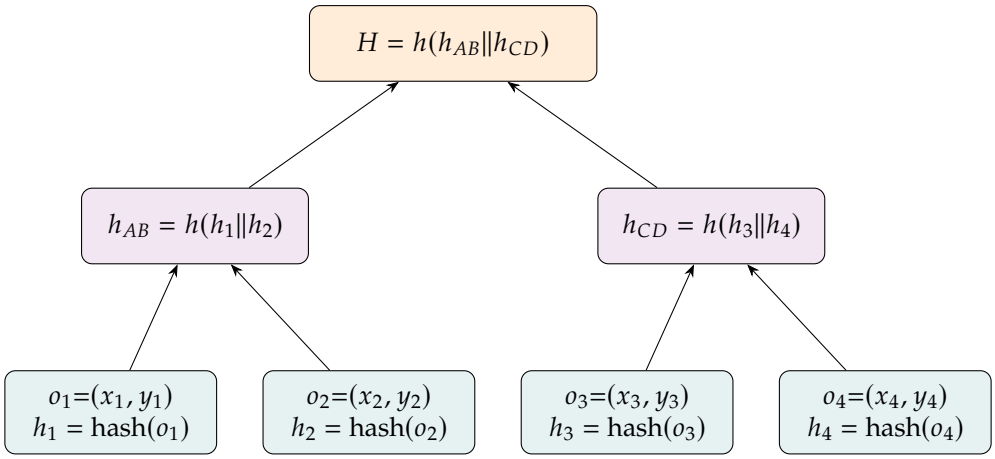


Figure 1: Merkle tree over an agent's observation set. Each leaf hashes one observation; parent hashes combine children; the root hash H fingerprints the whole set.

Sync. When agents i, j meet, they compare root hashes first:

$$H_i = H_j \;\Rightarrow\; O_i = O_j \quad \text{(nothing to send)},$$

$$H_i \neq H_j \;\Rightarrow\; \text{compare children, recurse only where hashes differ.}$$

This descends only into mismatched branches, so only the observations that actually differ are exchanged.

Why this helps. For a balanced tree over n observations, this takes

$$O(\log n) \quad \text{comparisons, instead of} \quad O(n)$$

for a naive full exchange – bandwidth is spent only on genuinely new information.

Beyond the Local Effect

Right now, we only consider the local effect of visiting a cell. That is, if we send an honest robot to a candidate cell s, we estimate how much new information we obtain at that particular cell or how much the reconstruction loss at that cell will decrease.

However, consider a cell that has already been visited by one or more adversarial agents. If we send an honest robot to this cell, the direct information gain at s may not be particularly large because the cell already has several observations. Nevertheless, the new observation provides additional evidence about the reliability of the agents that previously reported at s. This evidence can change their trust values α_i.

Since those agents may have reported observations at many other cells, a change in the trust of an agent does not only affect the reconstruction at s; it can affect the reconstructed values, and hence the reconstruction error, at all cells previously visited by that agent. Therefore, visiting s can have a *global effect* that is not captured by the local information gain alone.

Thus, in addition to the direct local effect of visiting a cell, we need to account for the indirect global effect caused by the resulting changes in the trust values of the agents that reported there.

Define the total reconstruction loss over all cells as

$$L_t = \sum_s e_t^2(s), \qquad e_t(s) = M_t(s) - T(s).$$

Differentiating with respect to α_i and applying the chain rule gives

$$\frac{\partial L_t}{\partial \alpha_i} = 2 \sum_s e_t(s) \, \frac{\partial M_t(s)}{\partial \alpha_i}.$$

Since $M_t(s) = W_t(s)/U_t(s)$ with $W_t(s) = \sum_j \alpha_j r_j(s)$ and $U_t(s) = \sum_j \alpha_j$, the quotient rule gives $\partial W_t(s)/\partial \alpha_i = r_i(s)$ and $\partial U_t(s)/\partial \alpha_i = 1$. Therefore,

$$\begin{aligned}
\frac{\partial M_t(s)}{\partial \alpha_i} &= \frac{U_t(s)\frac{\partial W_t(s)}{\partial \alpha_i} - W_t(s)\frac{\partial U_t(s)}{\partial \alpha_i}}{U_t^2(s)} \\
&= \frac{U_t(s) r_i(s) - W_t(s)}{U_t^2(s)} \\
&= \frac{U_t(s) r_i(s) - U_t(s) M_t(s)}{U_t^2(s)} \qquad (W_t(s) = U_t(s) M_t(s)) \\
&= \frac{U_t(s)\,[r_i(s) - M_t(s)]}{U_t^2(s)} \\
&= \boxed{\frac{r_i(s) - M_t(s)}{U_t(s)}}.
\end{aligned}$$

This quantifies how strongly agent i can shift the reconstructed value at s: it scales with agent i's disagreement from the current estimate, $r_i(s) - M_t(s)$, and is damped by the total trust mass $U_t(s)$ already supporting that cell.

Substituting back,

$$\frac{\partial L_t}{\partial \alpha_i} = 2 \sum_{s \in S_i} e_t(s) \, \frac{r_i(s) - M_t(s)}{U_t(s)},$$

where S_i is the set of cells reported by agent i. So, the *global leverage* of agent i is

$$\boxed{\Lambda_i = 2 \sum_{s \in S_i} e_t(s) \, \frac{r_i(s) - M_t(s)}{U_t(s)}}$$

Λ_i combines how strongly agent i can move each cell it has reported with how much that cell currently contributes to the loss, summed over all cells it has touched.

Expected Trust Change from Visiting a Cell

Visiting a candidate cell s may generate a cross-check that shifts the trust of any agent $i \in R_s$ that reported on it, by an expected amount $\Delta\alpha_i(s)$. A first-order Taylor expansion gives the resulting change in global loss,

$$\Delta L(s) \approx \sum_{i \in R_s} \Lambda_i \, \Delta\alpha_i(s),$$

so the expected *gain* from this indirect, trust-mediated effect is

$$\boxed{G_{\text{global}}(s) = - \sum_{i \in R_s} \Lambda_i \, \Delta\alpha_i(s)}$$

Updated Exploration Target

Adding this global term to the direct information-gain target $y_t(s)$ derived earlier gives the updated planning target

$$\boxed{y_t^{\text{new}}(s) = e_t^2(s) \, \frac{2U_t(s) + 1}{(U_t(s) + 1)^2} \; - \; \sum_{i \in R_s} \Lambda_i \, \Delta\alpha_i(s)}$$

The first term captures the direct reduction in error at s from an additional honest observation; the second captures the indirect benefit of visiting s through the trust updates it triggers for the agents that reported there.

Predicted Global Effect

The unknown trust change $\Delta\alpha_i(s)$ can be estimated by simulating the effect of a hypothetical observation at candidate cell s. The error-prediction network provides an estimate of the current squared error and its direction:

$$\hat{e}_t^2(s) = g_\phi(X_t),$$

$$\hat{q}_t(s) = \begin{cases} +1, & M_t(s) > T(s), \\ -1, & M_t(s) < T(s). \end{cases}$$

Since $T(s)$ is unknown, the predicted observation is obtained as

$$\hat{T}(s) = M_t(s) - \hat{q}_t(s)\sqrt{\hat{e}_t^2(s)}.$$

This predicted observation is then used to construct the hypothetical updated trust features for every agent $i \in R_s$ that has previously reported cell s. For each such agent,

$$c_i' = c_i + \mathbf{1}\left[|\hat{T}(s) - r_i(s)| > \delta_c\right],$$

with the corresponding updated values D_i' and z_i' computed using the same definitions as in the trust model.

The updated trust value is then obtained using one forward pass of the trained trust model:

$$\alpha_i' = f_\theta(c_i', z_i', D_i').$$

Therefore, the predicted change in trust caused by visiting s is

$$\boxed{\Delta\alpha_i(s) = \alpha_i' - \alpha_i}$$

for each agent that reported cell s.

If $\Delta\alpha_i(s) \neq 0$, this change affects not only cell s but every cell previously reported by agent i. Using the predicted error $\hat{e}_t(u)$, the corresponding leverage is

$$\hat{\Lambda}_i(u) = \hat{e}_t(u)\frac{r_i(u) - M_t(u)}{U_t(u)}.$$

Hence, the total leverage of agent i over all cells it has reported is

$$\boxed{\hat{\Lambda}_i^{\text{total}} = \sum_{u \in S_i} \hat{\Lambda}_i(u)}$$

and the predicted global effect of visiting s becomes

$$\boxed{\widehat{G}_{\text{global}}(s) = -\sum_{i \in R_s} \hat{\Lambda}_i^{\text{total}} \Delta\alpha_i(s)}$$

where only agents with a non-zero predicted trust change contribute to the sum.

This provides an estimate of the indirect global benefit of visiting s without requiring the actual future observation or the actual future trust values.

Planner

- The planner should direct agents toward the most informative locations, prioritising cells that are expected to provide the largest reduction in the current reconstruction uncertainty or error.
- The planner should ensure sufficient spatial overlap between agents so that independent reports can be compared and disagreement can be accumulated for reliable trust estimation. However, cross-over should not dominate the exploration process or significantly compromise spatial coverage.
- The planner should determine when further exploration is no longer sufficiently informative and terminate the exploration process once the expected gain from additional measurements becomes sufficiently small.
- As sufficient evidence is accumulated and the estimated trust values α_i become reliable, the planner should identify agents whose behavior is consistently inconsistent with the trusted population and remove or quarantine them from subsequent exploration.